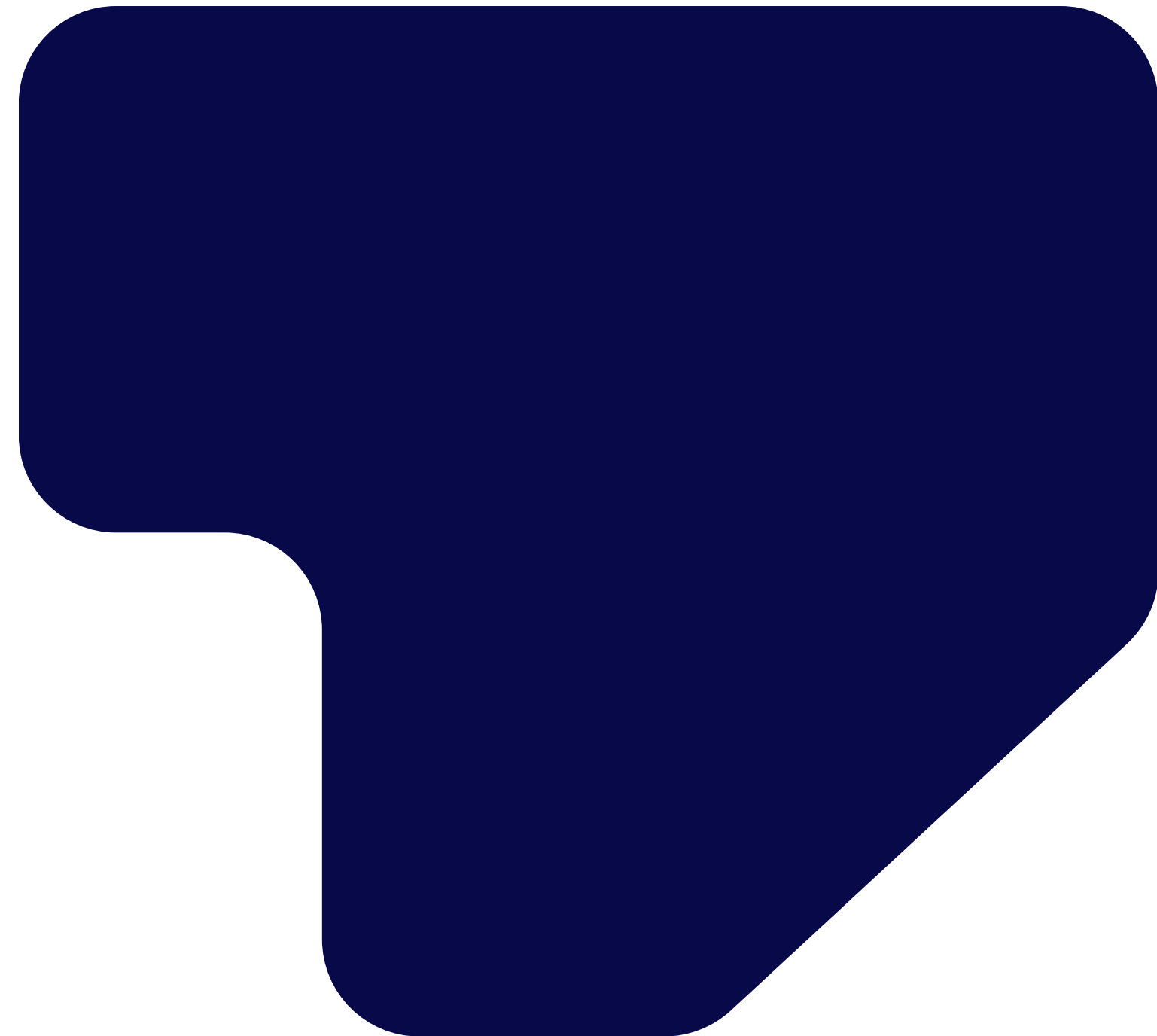




VidToCode

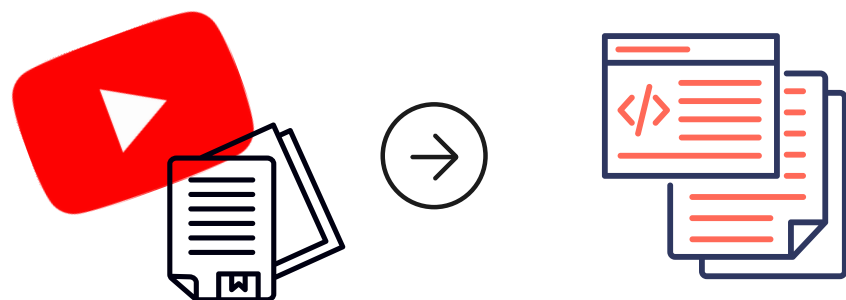
Ahmed Bahaj, Abdullah Misar, Yazeed Bafaqih, Albaraa Alnahari





Project Recap

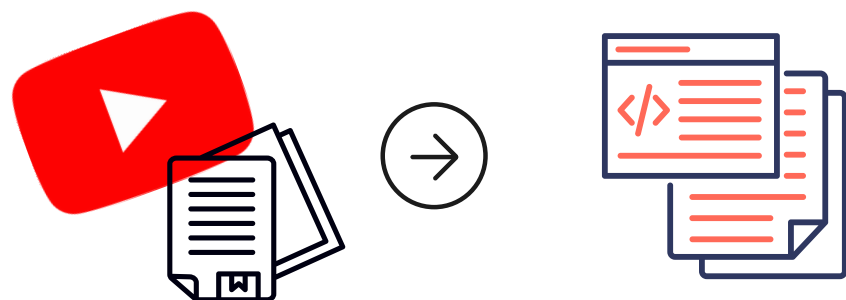
Goal





Project Recap

Goal



Novelty

- First system to use spoken narration and converts it to code



Project Recap

Goal



Novelty

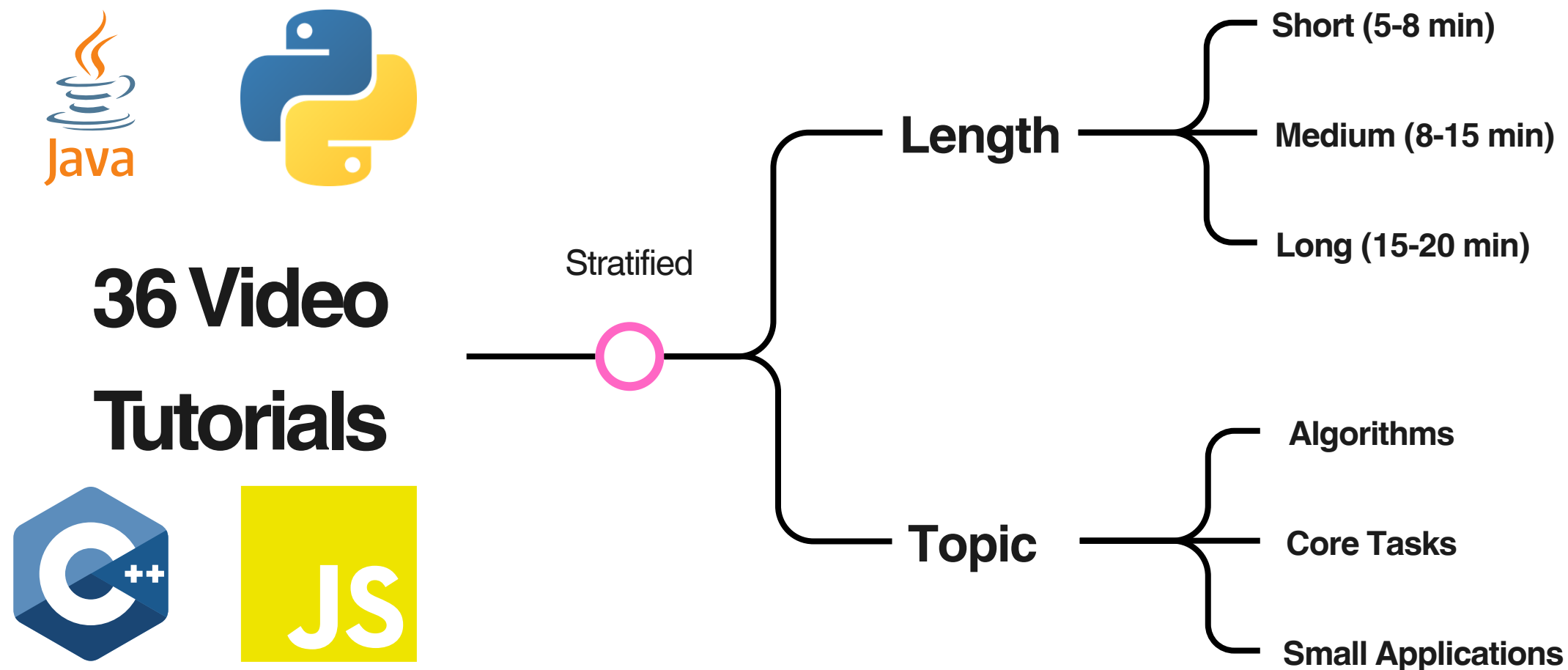
- First system to use spoken narration and converts it to code

Project Plan

- Phases:
 - **Phase 1: Literature Review (completed)**
 - **Phase 2: Project Plan (current)**
 - **Phase 3: Full Implementation & Evaluation (future)**

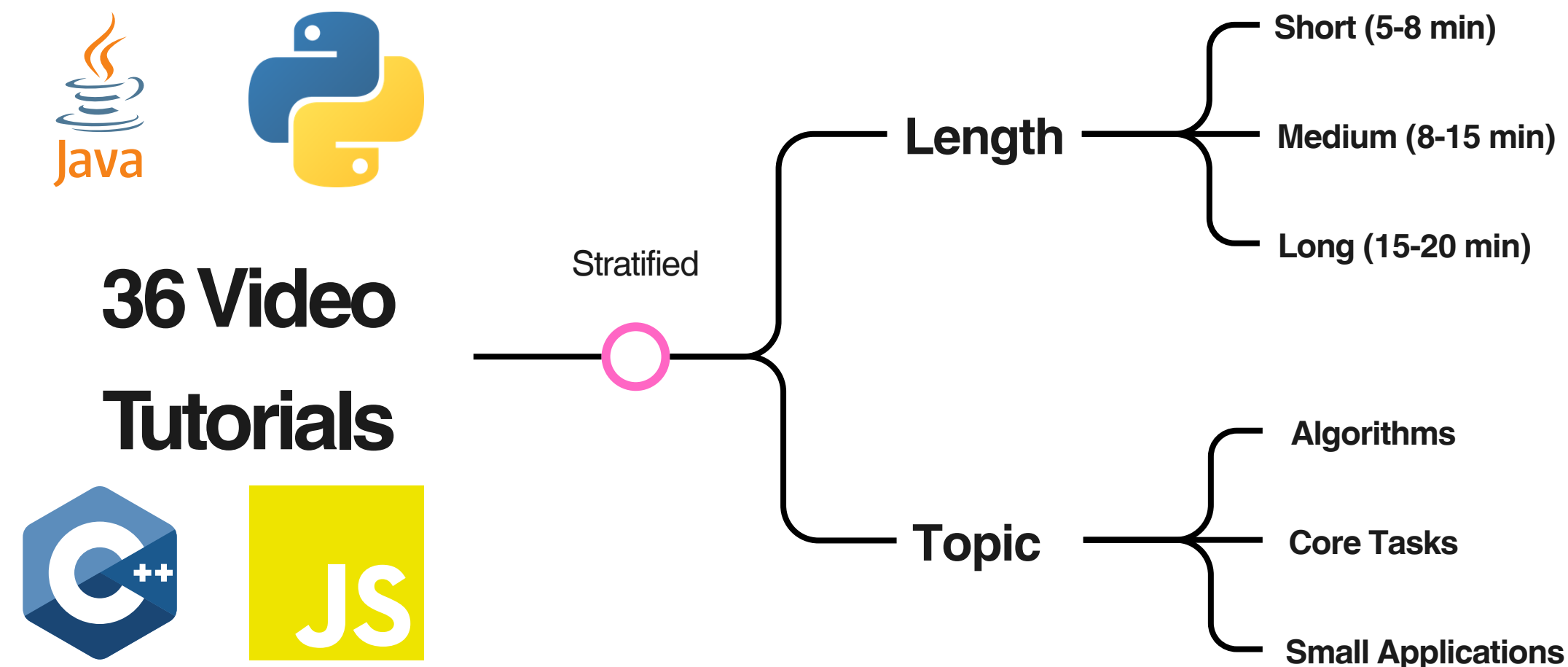


Phase 2.1 Dataset Collection





Phase 2.1 Dataset Collection



Per Video:

- Clean Transcript
- Final runnable src code
- Output in .txt
- Metadata (creator, duration, ID, title)

+7 Hours

Worth of tutorials were collected, curated, validated and cross-checked by teammates



Phase 2.2 Preprocessing & Annotation

1. Transcript Cleanning

- Remove filter words (“um”, “okay”)
- Remove off-topic phrases (“subscribe”)
- Normalize punctuation & casing
- Segment clips



Phase 2.2 Preprocessing & Annotation

1. Transcript Clearning

- Remove filter words (“um”, “okay”)
- Remove off-topic phrases (“subscribe”)
- Normalize punctuation & casing
- Segment clips

2. Pedagogical Tagging

Label narration segments by intent

- Implementation
- Explanation
- Debugging
- Refactoring



Phase 2.2 Preprocessing & Annotation



Before

"Okay, so now I'm gonna... um... add a
for loop here to iterate over the list..."

[raw transcript]

No structure
No programmatic semantic
Noise text



Phase 2.2 Preprocessing & Annotation



Before

"Okay, so now I'm gonna... um... add a for loop here to iterate over the list..."

[raw transcript]

No structure
No programmatic semantic
Noise text



After

```
{  
  "narration": "add a for loop to iterate over the list",  
  "intent": "implementation",  
  "code_snippet": "for item in my_list:",  
  "line_range": [12, 12]  
}
```

[structured data]

Aligned
Clean
Machine-friendly



Phase 2.3 Baseline Architecture

Prompt template:
Convert the following (narrations + source
code) into a concise docstring suitable for
code generation

{narration}



Teacher

zero-shot **docstring**
generation with a fixed
prompt template

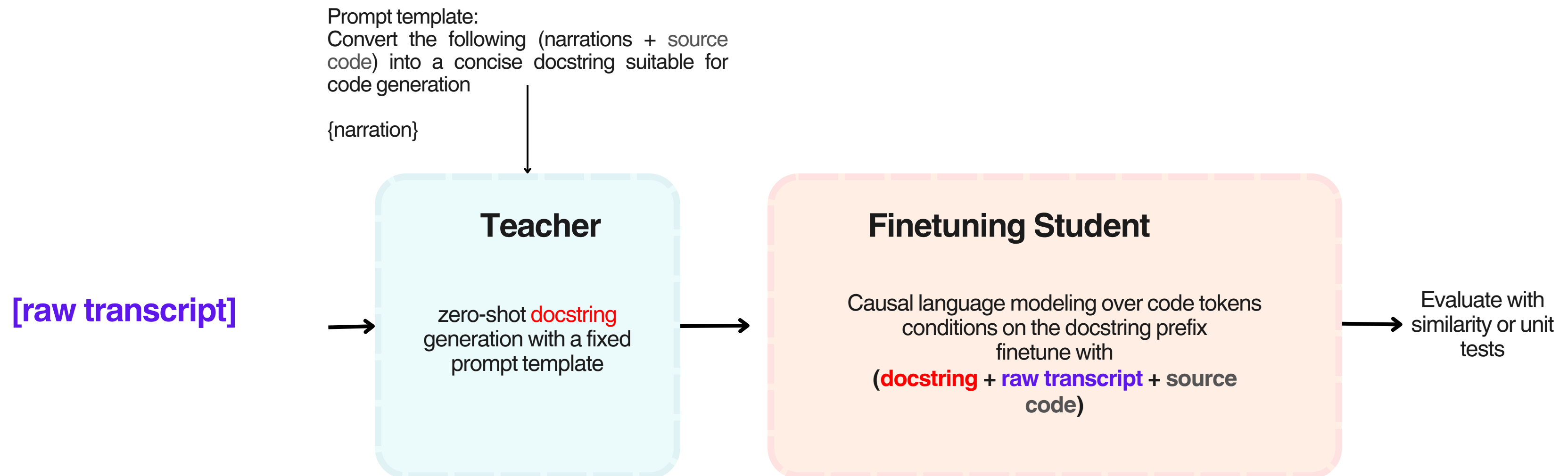


[raw transcript]



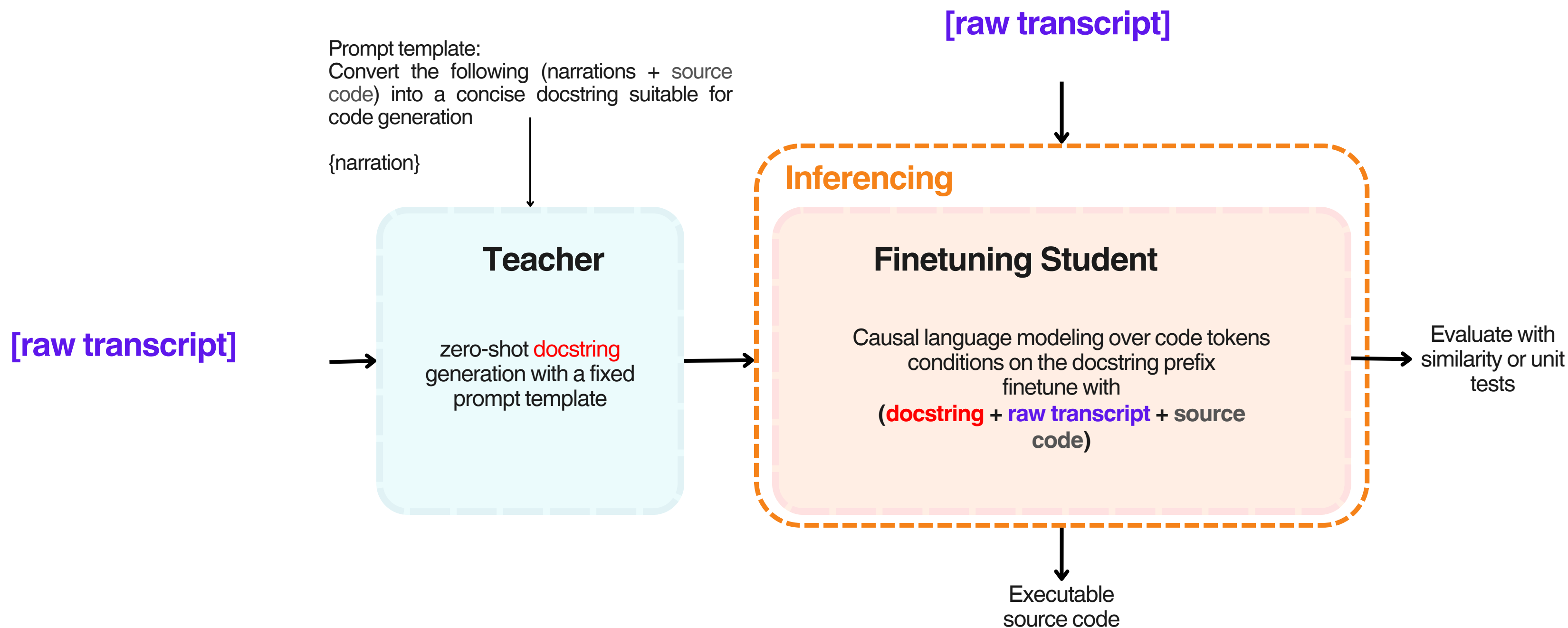


Phase 2.3 Baseline Architecture





Phase 2.3 Baseline Architecture





Phase 2.3 Baseline Architecture



- Sizes: 1B/ 3.7B/ 7B/ 16B
- Supports causal generation + infill

Pros

- Lightweight versions
- Purpose-built for code

Cons

- Best when code input already exists (docstring)

Remember: we aim to transcript → code



Phase 2.3 Baseline Architecture



- Sizes: 1B/ 3.7B/ 7B/ 16B
- Supports causal generation + infill

Pros

- Lightweight versions
- Purpose-built for code

Cons

- Best when code input already exists (docstring)

Remember: we aim to transcript → code

Qwen3.5-27B-Claude-4.6-Opus-Reasoning-Distilled-v2



- Sizes: 27B (MLX 4-bit)
- Supports causal generation + infill

Pros

- Distilled from Claude Opus 4.6's reasoning, it learned to think through natural language before generation.

Cons

- Heavy



Phase 2.3 Baseline Architecture

Pros

- Novel, frame-free. (Only textual transcript)
- Aligned with CodeGen2's training (docstrings)





Phase 2.3 Baseline Architecture

Pros

- Novel, frame-free. (Only textual transcript)
- Aligned with CodeGen2's training (docstrings)

Cons

- Limited generalization from small data
 - ~36 examples might overfit the approach
- Docstring conversion is non-trivial



Thank You